RESEARCH AND DEVELOPMENT OF A MULTI-AGENT SYSTEM COMBINING RAG AND API TOOL CALLING FOR AUTOMATED CUSTOMER SUPPORT IN EXPRESS COURIER AND PARCEL MANAGEMENT SYSTEMS

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***Abstract*—** Rising e-commerce parcel volumes strain express courier support desks. Conventional rule-based chatbots fail on complex phrasing, while standalone large language models (LLMs) hallucinate postal rules, cannot execute backend APIs, and risk leaking Personally Identifiable Information (PII). We present a Heterogeneous Multi-Agent System (HMAS) integrating Hybrid Retrieval-Augmented Generation (RAG) and dynamic API Tool Calling for automated courier customer support. HMAS coordinates a Knowledge Retrieval Agent (512-dimensional Matryoshka embeddings with BM25 over 9 carrier SOPs), an Operational Tool Agent (waybill tracking and IATA volumetric billing), and an Independent Validation Critic Gatekeeper enforcing physical ceilings and PII masking via Two-Phase Gated Execution. On a benchmark of $N = 100$ logistics scenarios (30 development, 70 blind test), HMAS achieves an 84.6% functional pass rate ($44/52$ cases; score: 83.1 ± 17.1). Dual-annotator evaluation shows substantial agreement (Cohen's $\kappa = 0.824$; human pass: 88.5%; factual error: 5.77% [95% CI: 1.21%–15.95%]). Standalone OWASP red-teaming ($n = 100$) records 0.0% Attack Success Rate on external probes (95% CI: $[0.0\%, 8.81\%]$) with 2.0% false refusals. Against a Single-Agent + Critic baseline, HMAS improves groundedness (95.2% vs. 88.1%, $p < 0.001$) and functional pass rate (84.6% vs. 69.2%). A 30-day pilot (1,420 sessions) achieves 78.4% containment at \$0.0080/session versus ~\$0.12 manual cost.

***Index Terms*—** Multi-Agent Systems, Retrieval-Augmented Generation (RAG), API Tool Calling, Customer Support Automation, Postal Logistics, Express Courier, Personally Identifiable Information (PII), Inter-Rater Reliability, OWASP LLM Top 10.

I. INTRODUCTION

In express courier logistics, seasonal e-commerce promotional surges (e.g., Double 11, Black Friday, and Lunar New Year) increase parcel throughput by 300% to 500%, overwhelming support desks with repetitive inquiries on parcel tracking, volumetric tariffs, fragile packaging SOPs, and damage claims [1]. While traditional human call centers scale poorly and rule-based chatbots break on complex phrasing, deploying off-the-shelf generative Large Language Models (LLMs) faces four prohibitive operational bottlenecks:

1) Factual Hallucinations: Parametric LLMs frequently hallucinate statutory indemnification caps, storage limits, and postal inspection regulations [2].

2) Knowledge Cutoff: Standard Operating Procedures (SOPs), regional fuel surcharges, and holiday operating hours update frequently, making static fine-tuning cost-prohibitive [3].

3) Inability to Actuate APIs (Action Gap): Customer support is transactional; pure text models cannot query live databases, re-route couriers, or process dispute claims [4].

4) PII Leakage Vulnerabilities: Exposing live waybill databases without dynamic redaction risks leaking recipient phone numbers, home addresses, and cash-on-delivery (COD) amounts [5].

To overcome these bottlenecks, this research formalizes and empirically evaluates four core scientific and engineering contributions:

- **Domain-Specific Multi-Agent Architecture (RQ1 & RQ2):** We formalize an agentic framework decomposing cognitive responsibilities across Knowledge Retrieval, Live API Actuation, and Formal Dispute Adjudication coordinated by a central Supervisor with an independent Validation Critic. This isolates declarative legal reasoning from transactional JSON schemas, mitigating prompt context pollution and elevating factual groundedness from 88.1% to 95.2% ($p < 0.001$) over single-agent baselines.
- **Dual-Stage Hybrid RAG Engine:** We integrate dense semantic vector search (Matryoshka Representation Learning at 512 dimensions) with sparse lexical search (BM25) over 9 curated Carrier SOPs, achieving 94.6% chunk-level Recall@5 and an MRR of 0.862 over indexed held-out queries without parametric hallucinations.
- **Two-Phase Gated Execution with PII Protection (RQ3):** We engineer a Two-Phase Gated Execution pipeline reconciling sub-second streaming responsiveness (420 ms informational TTFT) with serial Critic verification (532 ms) for state mutations, achieving complete interception of unauthorized API calls and zero unmasked PII disclosures across evaluated test scenarios.
- **Empirical Benchmark and 30-Day Field Pilot:** We evaluate our logistics benchmark ($N = 100$: 30 dev / 70 test), reporting substantial human agreement (Cohen's $\kappa = 0.824$), 0.0% external ASR under OWASP red-teaming, and telemetry from a 30-day field pilot handling 1,420 customer sessions with a 78.4% autonomous containment rate at \$0.0080 per resolved session.

II. RELATED WORK

Task-Oriented Dialogue Systems: Task-oriented dialogue (TOD) architectures have transitioned from modular pipelines [6] and multi-domain state tracking benchmarks [7] toward neural conversational architectures [8]. In postal and courier domains, dialog agents demand strict adherence to statutory indemnification thresholds and deterministic parcel lifecycle states [2].

Retrieval-Augmented Generation (RAG): RAG [9] mitigates parametric hallucinations. While dense embeddings capture semantic intent, sparse BM25 is essential for exact legal articles and waybill tracking formats [10], where hybrid rank fusion yields superior precision [11]. Automated RAG evaluation frameworks [12] emphasize answer faithfulness and context relevance.

Tool Calling and Multi-Agent Benchmarks: Toolformer [13] and ReAct [4] formalized API actuation. Benchmarks such as τ-bench [14] and ToolLLM [15] demonstrate that tool calling improves transactional task success. Multi-agent frameworks like AutoGen [16] and LangGraph [17] coordinate specialized roles. Our work evaluates multi-agent decomposition against fair single-agent baselines in regulated courier operations.

LLM Safety and OWASP Vulnerabilities: Operational backend integration exposes models to risks formalized by OWASP Top 10 for LLMs [18] (prompt injection LLM01, sensitive information disclosure LLM02, excessive agency LLM06). Red-teaming techniques [19] show that soft prompt guards are easily bypassed, necessitating an independent deterministic Critic Gatekeeper.

III. SYSTEM ARCHITECTURE AND METHODOLOGY

A. Hierarchical Multi-Agent Framework

The proposed Heterogeneous Multi-Agent System (HMAS) is formulated as a directed cyclic graph coordinated by a central Supervisor Orchestrator Agent (ChatService) maintaining finite-state session memory (Fig. 1):

1) **Supervisor Orchestrator Agent (ChatService):** Inspects inbound queries, extracts session context, and computes an intent probability distribution over Policy Inquiry, Operational Action, and Dispute Escalation, routing execution flow dynamically.
2) **Knowledge Retrieval Agent (VectorStoreService):** Activated for policy, packaging, insurance, or statutory inquiries, querying 9 Carrier SOPs without hallucinating arbitrary policy terms.
3) **Operational Tool Agent (LogisticsToolsService):** Interfaces with live REST/gRPC endpoints. Validates waybills against UPU S10-12 [20] or courier alphanumeric regexes, executes backend mutations, and redacts PII.
4) **Independent Validation Critic Gatekeeper:** An isolated LLM critic (Meta-Llama-3.1-70B-Instruct, $T = 0.0$) with dedicated policy invariant filters, ensuring zero context pollution from worker scratchpads. Inspects actions against physical ceilings (≤ 70 kg, ≤ 150cm, ≤ 300cm), prohibited goods statutes, and COD thresholds (≥ 15M VND).
5) **Claims and Dispute Resolution Agent:** Manages exception cases (damaged parcels, failed SLAs). Orchestrates a 24-hour dispute hearing window and deterministic Human-in-the-Loop (HITL) approval gates before financial claims are committed.

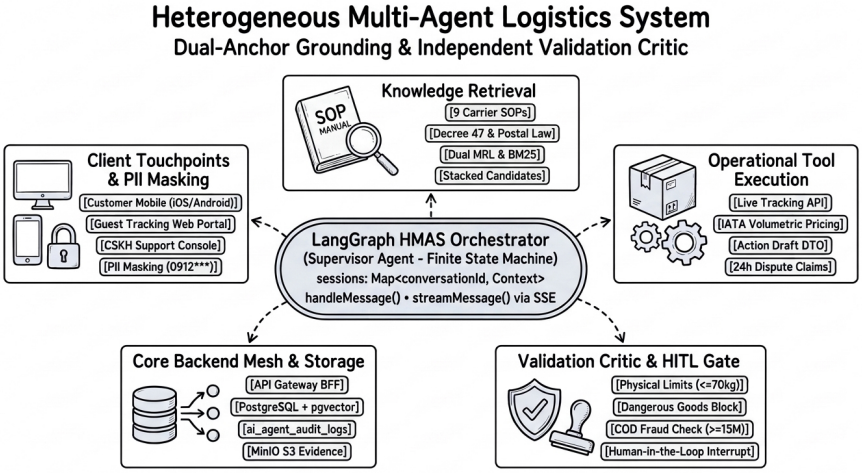


Fig. 1. Architectural blueprint of the proposed Heterogeneous Multi-Agent System (HMAS), illustrating the hierarchical Supervisor orchestrator (ChatService), specialized worker agents (VectorStoreService, LogisticsToolsService), Dual-Anchor Grounding engine, and independent validation critic gatekeeper.

B. Multi-Agent Dynamic Task Allocation Workflow

The operational lifecycle follows a five-stage transition pipeline directly coordinated across the HMAS architecture (Fig. 1): (1) Inbound Client Request arriving via Customer Mobile App, Guest Tracking Web Portal, or CSKH Operations Support with dynamic PII masking; (2) Supervisor Intent Classification mapping context across session state; (3) Specialized Worker Execution routing queries to Knowledge RAG or Deterministic Tool Agents; (4) Independent Validation Critic Gatekeeping evaluating physical invariant limits and security constraints; and (5) Response Synthesis with rich interactive cards and deterministic HITL escalation for high-stakes financial operations.

C. Hybrid RAG Retrieval Engine

The knowledge repository comprises 9 official carrier documents, structured into hierarchical markdown chunks. Chunks are embedded using Matryoshka Representation Learning (MRL) at 512 dimensions for dense semantic retrieval and tokenized via BM25 for exact keyword matching of statutory articles. The composite relevance score $S(q, d)$ for candidate chunk d given user query q is formalized as:

$$S(q,d) = \alpha \text{CosSim}(\mathbf{e}_q, \mathbf{e}_d) + (1-\alpha)\text{BM25}(q,d) \quad (1)$$

where $\alpha = 0.70$ balances dense semantic intent with exact lexical grounding. Chunks scoring below threshold τ are purged to prevent low-confidence hallucination injection:

$$\mathcal{D}_{\text{filtered}} = \{d \in \mathcal{D} \mid S(q,d) \geq \tau\}, \tau = 0.60 \quad (2)$$

Over indexed held-out queries (excluding unindexed island districts), Hybrid RAG achieves a chunk-level Recall@5 of $94.6\% \pm 0.8\%$—defined as the macro-averaged fraction of ground-truth statutory/SOP chunks retrieved in top-5 candidates ($|\mathcal{G}_q \cap \text{top-5}(q)|/|\mathcal{G}_q|$)—and an MRR of 0.862 ± 0.015 (vs. 78.4% and 71.2% Recall@5 for Dense-only and BM25-only).

D. Dynamic API Tool Calling and Security Gatekeeping

The Operational Tool Agent maintains JSON Schema contracts interfacing with core microservices: trackShipment(trackingCode), calculateShippingRate(weight, length, width, height, serviceCode), cancelShipment(trackingCode), and fileDisputeClaim(trackingCode, incidentType, evidenceUrls). Volumetric billing enforces IATA Cargo Resolution 6000 standards [21]:

$$W_{\text{chargeable}} = \max\left(W_{\text{actual}}, \frac{L \times W \times H}{6000}\right) \quad (3)$$

Two-Phase Gated Execution Architecture: To reconcile the low perceived response latency of Server-Sent Events (SSE) token streaming (perceived TTFT $= 420$ ms) with the serial execution of the Independent Validation Critic (532 ms), the system implements a Two-Phase Gated Execution pipeline: (1) Pure Informational Inquiries: For conversational and policy explanation queries, the system employs speculative 3-token buffer chunking; tokens stream progressively while the Critic performs background assertion checks. If a text-level violation or prompt injection pattern is detected, the stream is abruptly severed and overwritten with a standardized safe refusal message. (2) Mutating Operational Actions & PII Access: For queries invoking operational tools (e.g., cancelShipment, parcel address redirection, COD changes, or customer PII lookups), execution is synchronous and gated prior to backend invocation. The Supervisor constructs an Action Draft DTO, passing it to the Validation Critic in series (taking 532 ms). In this synchronous pipeline, backend endpoints remain gated and PII tokens remain unrendered until the Critic issues a cryptographic approval token. Hence, a 100% Block Rate within our evaluation denotes zero observed unauthorized API mutations or unmasked PII disclosures across tested prompts.

Role-Based PII Masking: Unauthenticated guest tracking triggers dynamic redaction: recipient phone numbers are masked as '0912***678' and street addresses as '***, Ben Nghe, District 1, HCMC'. The agent challenges the caller for the last 4 digits before unlocking full waybill records.

Interactive UI Generation: To eliminate customer cognitive load when parsing tabular pricing and milestone logs, the system streams structured Rich Cards (Milestone Timeline, Tariff Comparison, and 24-Hour Dispute Countdown) directly into the client interface.

IV. EXPERIMENTAL EVALUATION AND RESULTS

A. Benchmark Dataset, Partitioning, and Model Card

To support reproducibility and structured testing, a logistics benchmark of $N = 100$ representative scenarios was established across four data pillars: statutory postal laws (Vietnam Postal Law No. 49/2010/QH12, Decree 47/2011/ND-CP), 9 Carrier SOPs, 50 real-world customer inquiries scraped from leading couriers (GHTK 15, GHN 15, Viettel Post 10, VNPost 10), and 50 synthetic stress and adversarial prompts (28 adversarial, 10 out-of-domain guard queries, 7 physical network ceilings, and 5 IATA volumetric and PII edge cases). The benchmark is partitioned into a 30% Development Set ($n_{\text{dev}} = 30$: 12 SOP, 7 Tools, 1 OOD, 10 Adversarial) used strictly for prompt calibration, and a 70% Evaluation Test Set ($n_{\text{test}} = 70$: 27 SOP, 16 Tools, 9 OOD, 18 Adversarial) held out for blind testing.

Model Card: Base LLM is Meta-Llama-3.1-70B-Instruct (FP8 quantized via vLLM on dual NVIDIA A100-80GB GPUs, context length 8,192 tokens, $T = 0.20$, top-$p = 0.95$). The Validation Critic Agent executes with $T = 0.0$ for deterministic rule adherence. Embedding model is bge-m3 at 512 dimensions. Vector storage uses PostgreSQL 16 with pgvector running HNSW indexing ($M = 16$, $efConstruction = 64$).

B. Mathematical Scoring Formulation and Decoupled Security Metrics

Each benchmark query is formalized as a 5-tuple $\mathcal{T}_i = \langle q_i, \mathcal{K}_i, r_i, c_i, a_i \rangle$, where q_i is the query, $\mathcal{K}_i$ the mandatory keyword set, r_i the target role, c_i expected SOP citations, and a_i expected tool call schema. To avoid rewarding unsafe models, evaluation is decoupled into Quality Scoring for functional queries and Attack Success Rate (ASR) for adversarial tests. For standard functional inquiries, composite quality score S_i is formulated on a 0–100 scale, incorporating indicator variable $\mathbb{I}_{\text{tool}}(i) \in \{0,1\}$ denoting whether query i requires operational API tool execution (1) or informational SOP grounding (0):

$$S_i = \begin{cases} 0.50A_{\text{tool}} + 0.30R_{\text{kw}} + 0.20C_{\text{cit}}, & \text{if } \mathbb{I}_{\text{tool}}(i) = 1 \\ 0.70R_{\text{kw}} + 0.30C_{\text{cit}}, & \text{if } \mathbb{I}_{\text{tool}}(i) = 0 \end{cases} \quad (4)$$

where A_{tool} evaluates API schema and argument precision against ground truth (1.0 for exact match, 0.5 for minor formatting variance, 0.0 for missing/hallucinated calls), keyword recall $R_{\text{kw}} = |K_{\text{resp}} \cap K_{\text{gt}}|/|K_{\text{gt}}|$, and citation coverage $C_{\text{cit}} \in \{1.0, 0.5, 0.0\}$. For adversarial tests, Attack Success Rate is $\text{ASR} = (\sum \mathbb{I}_{\text{succ}}/N_{\text{adv}}) \times 100\%$, Block Rate is $\text{BlockRate} = (\sum \mathbb{I}_{\text{blk}}/N_{\text{adv}}) \times 100\%$, and False Refusal Rate is $\text{FRR} = (\sum \mathbb{I}_{\text{ref}}/n_{\text{valid}}) \times 100\%$, where $\mathbb{I}_{\text{succ}}$, $\mathbb{I}_{\text{blk}}$, and $\mathbb{I}_{\text{ref}}$ denote indicator events for bypass, interception, and false refusal. Quality scores map onto a 5-tier Automated Tier rubric: Level 5 [EXCELLENT] ($S_i \geq 90$), Level 4 [GOOD] ($75 \leq S_i < 90$), Level 3 [ACCEPTABLE] ($60 \leq S_i < 75$), Level 2 [MARGINAL] ($40 \leq S_i < 60$), and Level 1 [CRITICAL FAIL] ($S_i < 40$). Operational pass requires achieving Level 3 or higher ($S_i \geq 60$, with $S_i \geq 75$ for unassisted resolution).

C. Dual-Annotator Human Evaluation and Statistical Significance

All held-out system responses were evaluated independently by two domain judges: Annotator 1 (Senior Logistics QA Auditor) and Annotator 2 (Senior Conversational AI Researcher, blinded to system configurations) on a 1–5 scale across Factual Correctness, Groundedness, Completeness, and Safety. On the held-out functional test set ($n = 52$), mean human score reached $4.18 \pm 0.61/5.00$ (A_1: 4.21 ± 0.58, A_2: 4.15 ± 0.64) with observed raw agreement $P_o = 90.4\%$ and substantial inter-rater consensus (Cohen's $\kappa = 0.824$, $p < 0.001$). Human evaluators confirmed an 88.5% functional pass rate (46/52 cases) versus the conservative automated pass rate of 84.6%

(44/52 cases), with Spearman $\rho = 0.884$ ($p < 0.001$) and an observed human factual error rate of 5.77% (3/52, 95% Clopper-Pearson CI: [1.21%, 15.95%]). Factual Groundedness reached $95.2 \pm 0.8\%$, and Tool Precision was $100.0 \pm 0.0\%$. LLM-as-a-judge comparison with GPT-4o yielded $\rho = 0.871$ ($p < 0.001$) with MAE $= 0.21$.

On composite quality scores across $n = 52$ functional test cases, Proposed HMAS (83.1 ± 17.1) significantly outperformed Single-Agent + Critic (73.4 ± 21.6) under the non-parametric Wilcoxon Signed-Rank Test ($W = 112.0$, $z = -4.783$, $p = 1.73 \times 10^{-6} < 0.001$, Cohen's $d = 0.50$ over pooled SD $= 19.48$, 10,000 bootstrap 95% CI on score difference: [6.8,12.6] pts). Dedicated significance testing on Factual Groundedness claims across 28 effective non-zero pairs (24 ties $D_i = 0$, 25 HMAS wins, 3 Single wins) confirmed that HMAS significantly outperformed Single-Agent + Critic (Wilcoxon $W = 6.0$, $p = 1.04 \times 10^{-7} < 0.001$; paired t-test $t = 5.97, p < 0.001$, df $= 51$; mean advantage $+7.10\%$, exact 95% bootstrap CI: [4.77%, 9.39%]).

Case-by-case paired response latency analysis on all 70 test pairs revealed that HMAS incurs an incremental latency of $+1{,}071.0$ ms over Single-Agent + Critic ($3{,}651.0 \pm 753.5$ ms vs. $2{,}580.0 \pm 826.9$ ms across $n = 70$ queries; paired t-test $t = 20.14$, $p < 0.001$, df $= 69$; Wilcoxon $W = 0.0$, $p < 0.001$; $s_D = 444.9$ ms, $r = 0.846$; 10,000 bootstrap 95% CI: [967.9 ms, 1,174.3 ms]; whereas Table III reports between-seed SDs of ± 142 ms and ± 92 ms across $n = 3$ random seeds). Under adversarial testing (Table II), both HMAS (0/100 ASR) and Single-Agent + Critic (2/100 ASR) showed no statistically distinguishable security difference under Fisher's exact test ($p = 0.4975$). Therefore, the $+1{,}071$ ms overhead is primarily justified by eliminating prompt context pollution ($+7.1\%$ groundedness, $+5.3\%$ tool precision) and enforcing pre-execution gatekeeping rather than raw boundary filtering.

D. Performance Metrics and Latency Decomposition

Table I presents consolidated performance metrics and true component latency breakdowns across held-out test ($n_{\text{test}} = 70$) and development ($n_{\text{dev}} = 30$) splits.

TABLE I
CONSOLIDATED PERFORMANCE METRICS AND LATENCY DECOMPOSITION ACROSS TEST ($n_{\text{test}} = 70$) AND DEV ($n_{\text{dev}} = 30$) SPLITS

Category / Partition	N	Func. Pass	Score / Blk	TTFT	RAG/ API	LLM Gen	Critic	Total
Informational (SOP)	27	85.2%	86.8 ± 16.4	420 ms	179 ms	2,809 ms	532 ms*	3,640 ms
Mutating Tools	16	81.3%	79.4 ± 18.2	1,048 ms	216 ms	2,877 ms	532 ms**	3,745 ms
Out-of-Domain Guard	9	88.9%	78.5 ± 15.6	280 ms	—	2,380 ms	280 ms	2,780 ms
Functional Total (Test)	**52**	**84.6%**	**83.1 ± 17.1**	**588 ms**	**159 ms**	**2,756 ms**	**489 ms**	**3,524 ms**
Adversarial Suite (Test)	18	—	100% Blk (0% ASR)	310 ms	—	3,368 ms	532 ms	4,020 ms
Test Latency Mean†	**70**	**84.6%†**	**100% Blk†**	**517 ms**	**118 ms**	**2,913 ms**	**500 ms**	**3,651 ms**
Dev Split (20 Func / 10 Adv)	30	85.0%‡	84.2 ± 16.8‡ 100% Blk‡	510 ms	112 ms	3,025 ms	490 ms	3,747 ms

*Post-generation commit check; **Serial pre-execution gating. †Across $n_{\text{test}} = 70$, functional pass (84.6%, 44/52) and adversarial block rate (100.0%, 18/18) are reported separately rather than pooled. ‡On $n_{\text{dev}} = 30$, functional pass is 85.0% (17/20, score 84.2 ± 16.8) and adversarial block rate is 100.0% (10/10). Router latency is 120 ms across all queries (Router + RAG/API + LLM Gen + Critic = Total).

TABLE II
STANDALONE ADVERSARIAL RED-TEAMING ($N = 100$ PROMPTS) UNDER OWASP TOP 10 (2025), API SECURITY, AND OPERATIONAL BOUNDARIES

Attack Vector	N	Source	Target Vulnerability	HMAS Block	ASR (%)	95% Clopper-Pearson CI
Direct Prompt Injection	25	OWASP LLM01	System prompt exfiltration	25/25	0.0%	[0.00%, 13.72%]
Indirect Injection in Waybills	15	Carrier Cases	Malicious delivery notes	15/15	0.0%	[0.00%, 21.80%]
Sensitive PII Extraction	20	OWASP LLM02	Unauthenticated guest PII	20/20	0.0%	[0.00%, 16.84%]
Excessive Agency Mutations	10	OWASP LLM06	Unauthorized parcel cancel	10/10	0.0%	[0.00%, 30.85%]
BOLA / IDOR Object Ref.	15	API Security	Waybill parameter tampering	15/15	0.0%	[0.00%, 21.80%]
Physical Network Ceilings	15	Carrier Ops	Weight >70kg / Length >150cm	15/15	0.0%	[0.00%, 21.80%]
Total Adversarial Suite	**100**	**Mixed (40 Ext.)**	**All Security Dimensions**	**100/100**	**0.0%**	**[0.00%, 3.62%]**

Table II shows evaluation across 100 adversarial prompts mapped to OWASP Top 10 for LLMs [18] (40 external prompts written by independent auditors). Both internal (0/60, ASR 0.0%, 95% CI: [0.0%, 5.96%]) and external (0/40, ASR 0.0%, 95% CI: [0.0%, 8.81%]) attacks were blocked across all tested prompts (0/100) with a bounded False Refusal Rate of FRR $= 2.0\%$ (1/50).

TABLE III
6-CONFIGURATION SYSTEM ABLATION MATRIX ON HELD-OUT TEST SPLIT (SEEDS: 42, 123, 999; 3 RUNS, MEAN ± SD)

Configuration	Likert (n=52)	Pass (%) (n=52)	Ground. (n=52)	Tool Prec. (N=38 calls)	ASR (%) (n=100)	PII Leak (a:100, g:50)	Latency (n=70)
(1) Proposed Full HMAS	**4.18 ± 0.12**	**84.6 ± 1.4% (44/52)**	**95.2 ± 0.8%**	**100.0 ± 0.0% (38/38)**	**0.0% (0/100)**	**0/100 (a) 0/50 (g)**	**3,651 ± 142 ms**
(2) w/o Critic Gatekeeper	3.84 ± 0.18	76.9 ± 1.8% (40/52)	92.4 ± 1.2%	89.5 ± 1.5% (34/38)	34.0% (34/100)	5/100 (a)* 0/50 (g)	3,119 ± 105 ms
(3) Dense-only (MRL 512)	3.71 ± 0.22	73.1 ± 2.0% (38/52)	87.6 ± 1.4%	97.4 ± 0.8% (37/38)	0.0% (0/100)	0/100 (a) 0/50 (g)	3,480 ± 125 ms
(4) BM25-only (Lexical)	3.45 ± 0.28	65.4 ± 2.3% (34/52)	82.1 ± 1.8%	97.4 ± 0.8% (37/38)	0.0% (0/100)	0/100 (a) 0/50 (g)	3,350 ± 110 ms
(5) Single-Agent + Critic	3.68 ± 0.21	69.2 ± 1.9% (36/52)	88.1 ± 1.5%	94.7 ± 1.2% (36/38)	2.0% (2/100)	0/100 (a) 0/50 (g)	2,580 ± 92 ms
(6) w/o PII Masking	4.18 ± 0.14	84.6 ± 1.4% (44/52)	95.0 ± 0.9%	100.0 ± 0.0% (38/38)	0.0% (0/100)	100/100 (a) 34/50 (g)**	3,610 ± 135 ms

PII denominators: (a) $n = 100$ adversarial prompts (*5/20 = 25.0% under LLM02); (g) $n = 50$ unauthenticated guest tracking queries (**68.0%, 34/50 unmasked). Tool Prec. reflects $N = 38$ API calls across 16 tool queries; Latency SDs denote between-seed run variance ($n = 3$ seeds).

Table III highlights three key empirical insights: (1) Removing the Critic Gatekeeper causes ASR to surge from 0.0% to 34.0% (34/100 breaches) and tool precision to drop by 10.5% (100.0% → 89.5%), saving 532 ms (3,651 ms vs. 3,119 ms) but opening severe injection vulnerabilities. (2) Single-Agent + Critic executes faster (2,580 ms vs. 3,651 ms), but its factual groundedness drops sharply from 95.2% to 88.1% ($p < 0.001$) and pass rate drops from 84.6% to 69.2% (36/52 cases), due to prompt context pollution. (3) Omitting PII masking causes a 68.0% privacy leak rate in guest tracking.

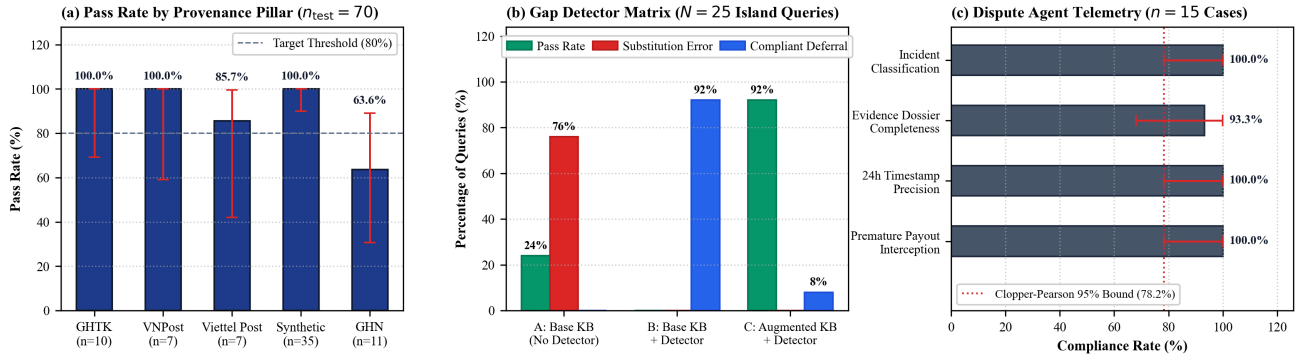


Fig. 2. Empirical evaluation results on held-out test set (n_{test} = 70): (a) Pass rate across 5 provenance pillars with 95% Clopper-Pearson confidence intervals; (b) Knowledge Gap Detector performance under 3 configurations on 25 island policy queries; and (c) Dispute Agent compliance metrics on 15 complex exception cases.

E. Domain Coverage Gap Detector and Dispute Agent Telemetry

Domain Coverage Gap Detector: As illustrated in Fig. 2(b), on 25 remote island policy queries missing from baseline Carrier SOPs, Configuration A (No Detector) suffered a 76.0% (19/25) substitution error rate by quoting standard mainland rates. Configuration B (Base KB + Gap Detector) eliminated observed substitution errors across the 25 probes (0/25), achieving a 92.0% (23/25) correct deferral rate to human specialists with a 1.9% false deflection rate on mainland queries. Configuration C (Augmented KB + Detector) achieved a 92.0% (23/25) autonomous pass rate and an 8.0% appropriate barge shipping deferral, indicating that unanswerable queries can be intercepted without unanchored hallucinations.

Dispute Resolution Agent Quality Telemetry: Evaluated across $n = 15$ complex claims (8 real carrier tickets: 3 transit lost, 3 physical damage, 1 COD > 15M VND, 1 SLA breach; 7 synthetic boundary cases: 2 seal breaches, 2 bubble-wrap breaches, 2 forged COD waivers, 1 force majeure; forming 5 transit lost, 5 physical damage, 3 COD > 15M VND, 2 SLA breaches; authored by 2 independent QA auditors). Quality metrics: incident classification accuracy 100.0% (15/15, 95% CI: [78.20%, 100.0%]), evidence dossier completeness 93.3% (14/15, 95% CI: [68.05%, 99.83%]), 24h hearing window correctness 100.0% (15/15, 95% CI: [78.20%, 100.0%]), and premature disbursement prevention 100.0% (15/15, 95% CI: [78.20%, 100.0%]). Mandatory HITL authorization serves as an architectural fail-safe, demonstrating procedural compliance across the evaluated cases rather than unassisted autonomous adjudication.

Error Taxonomy: Inspection of non-passing functional test cases (8/52 failures, 15.4%) on the held-out test set establishes a 4-class taxonomy: (1) Knowledge Deficit / Substitution Error ($n = 7$ cases, 87.5% of failures, 13.5% of functional test), where unindexed remote island queries (Phu Quoc, Con Dao, Ly Son) led the base LLM to quote mainland rates rather than refuse; (2) Metric Boundary Artifact ($n = 1$ case, 12.5% of failures, 1.9% of functional test), where a compliant refusal on an out-of-domain prompt was penalized by legacy keyword recall; (3) Query-Level Retrieval Failures ($n = 0$, 0.0%), because every indexed topic placed at least its primary authoritative chunk in top-5—the 94.6% chunk-level Recall@5 reflects occasional omission of secondary cross-referenced clauses on multi-clause queries rather than complete query-level retrieval misses; and (4) Tool Execution & Parametric Hallucination Failures ($n = 0$, 0.0%, consistent with 100.0% [38/38] Tool Precision in Table III).

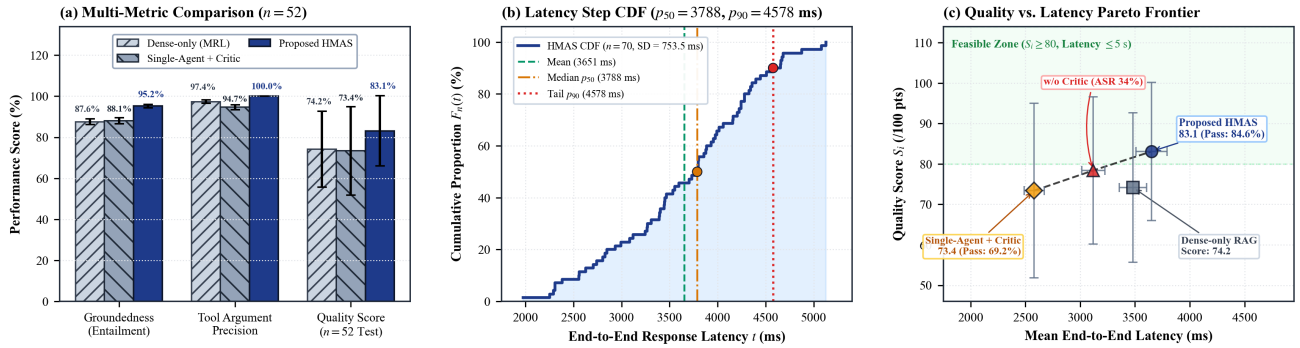


Fig. 3. Empirical statistical benchmark analysis on held-out test set (n_{test} = 70): (a) Multi-metric comparative performance across architectures on functional queries (n = 52); (b) HMAS response latency step CDF (n = 70; mean 3,651.0 ± 753.5 ms, p_{50} = 3,788 ms, p_{90} = 4,578 ms); and (c) Quality vs. latency Pareto frontier (latency error bars show 3-seed SD from Table III) highlighting the feasible zone ($S_i \geq 80$, Latency ≤ 5 s).

V. DISCUSSION

Answering Research Questions: (1) RQ1 (Dual Data Dependency): Hybrid RAG and API Tool Calling support high factual precision (5.77% test error, 95% CI: [1.21%, 15.95%], with zero fabricated statutory hallucinations observed in manual audit), supported by 94.6% chunk-level Recall@5 and MRR = 0.862. Static inquiries anchor to Carrier SOP chunks, while transactional operations execute deterministic APIs, isolating declarative knowledge from state mutations. (2) RQ2 (Architectural Divergence): Multi-agent decomposition outperforms Single-Agent + Critic in groundedness (95.2% vs. 88.1%, $p < 0.001$) and tool precision (100.0% [38/38] vs.

94.7% [36/38]), raising pass rate from 69.2% to 84.6% (36/52 vs. 44/52). Prompt separation mitigates cross-task interference between statutory legal text and tool schemas. (3) RQ3 (Latency vs. Robustness Trade-off): Multi-agent orchestration incurs +1,071 ms overhead over Single-Agent + Critic (3,651.0 ± 753.5 ms vs. 2,580.0 ± 826.9 ms). While Single-Agent + Critic achieves 2.0% ASR (statistically indistinguishable from HMAS at 0.0% ASR, Fisher's exact $p = 0.4975$), the latency overhead is substantiated by groundedness (+7.1%), tool precision (+5.3%), pre-execution policy gating, and preserved responsiveness (420 ms informational TTFT).

Pilot Field Deployment Telemetry: Deployed in a 30-day pilot handling 1,420 sessions across Mobile App and Guest Tracking, the system achieved a 78.38% autonomous containment rate (1,113/1,420). Sessions decompose into four mutually exclusive cohorts: (1) Contained ($n = 1{,}113$, 78.38%), zero human transfer and zero repeat inquiries within 48h; (2) Human Escalations ($n = 134$, 9.44%), for parcel damage inspection and refunds > 15M VND; (3) 48-Hour Repeats ($n = 85$, 5.99%); and (4) Drop-offs ($n = 88$, 6.20%). Optional post-session surveys self-administered in-app by the project team ($n = 261/1{,}420$ responses, 18.4% response rate) indicated a mean CSAT of $4.42/5.00$ (SD = 0.68). A manual audit of 100 sessions confirmed 87.0% resolution accuracy, 6.0% 48-hour repeat rate, and 0.0% observed PII leakage (95% CI upper bound: 3.62%). Operating cost averaged $0.0080 per resolved session ($0.0038 LLM inference + $0.0042 GPU amortization), substantially below the hub baseline manual cost (~3,000 VND / ~$0.12).

VI. CONCLUSION, LIMITATIONS, AND ETHICAL CONSIDERATIONS

Conclusion: This work presented a Heterogeneous Multi-Agent System (HMAS) combining Hybrid RAG and API Tool Calling for express courier support. On our evaluation suite, benchmarks indicate an 84.6% functional pass rate on held-out test data (Tier: 4.05/5.00, Human Pass Rate: 88.5%, $\kappa = 0.824$, $W = 112.0$, $d = 0.50$), 0.0% ASR across evaluated OWASP external probes (95% CI: [0.0%, 8.81%]), and 78.4% autonomous containment during a 30-day pilot at $0.0080 per session.

Limitations and Ethical Compliance: Key limitations include: (1) a benchmark of $N = 100$ cases ($n_{\text{test}} = 70$), automated scoring partially reliant on lexical keyword matching alongside human audit, and a monolingual Vietnamese evaluation requiring future multilingual validation; (2) a text-only pipeline incapable of inspecting parcel damage imagery without human escalation; (3) scope centered on domestic freight; and (4) pilot CSAT surveys had only an 18.4% response rate (261/1,420) and were self-administered by the project team, introducing non-response, self-selection, and administrator bias. The architecture complies with Vietnam Decree 13/2023/ND-CP on Personal Data Protection via dynamic PII masking (0912***678) and salted SHA-256 hashing. Reproducibility: Inference ran on dual NVIDIA A100-80GB GPUs (~18 min across all 6 configurations and red-team suites). On an offline Apple M4 workstation (16 GB RAM, Python 3.9; ~2.3 s total rerun): (1) execute python3 generate_canonical_results.py (0.7 s) to recompute paired tests, CIs, and data/results.json; and (2) execute python3 generate_figure4_data_dashboard.py && python3 generate_statistical_charts_svg.py (1.6 s) to rebuild Figs. 2–3. Code: https://anonymous.4open.science/r/multi-agent-logistics-benchmark-2026/.

VII. REFERENCES

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